Speak **L**ike **A M**athematician:

solve a right triangle:

inverse trig ratios:

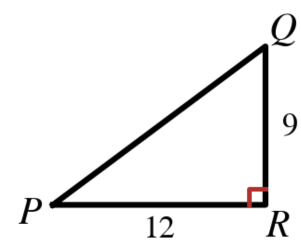
Inverse Trigonometric Ratios

θ = sin⁻¹(opposite / ____________) · θ = cos⁻¹(____________ / hypotenuse) · θ = tan⁻¹(opposite / ____________)

Plan for solving a triangle: find the third side with the ____________________ Theorem (or a trig ratio), find one acute angle with an ____________________ ratio, then subtract from ________° for the other.

Example 1 – Two Legs Known

In △PQR at the right, the right angle is at R, PR = 12 and QR = 9. Round decimals to the nearest tenth.



a) Find PQ.

b) Find m∠P.

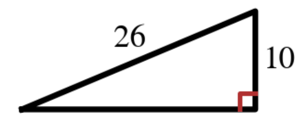
c) Find m∠Q.

Example 2 – Hypotenuse and a Leg

The right triangle at the right has hypotenuse 26 and a leg of 10.

a) Find the third side.

b) Find both acute angles.

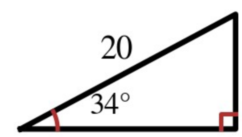


Example 3 – An Angle and the Hypotenuse

The triangle at the right has a 34° angle and hypotenuse 20.

a) Find both legs.

b) Find the other acute angle.



Example 4 – An Angle and a Leg

The triangle at the right has a 58° angle whose ADJACENT leg is 7.

a) Find the other leg and the hypotenuse.

b) Find the remaining angle, then list all six parts.